

MEWS Development Program

Software Lifecycle, Quality & Engineering Practices

Document Type: Portfolio Reference & Sample

Purpose: Demonstrate how software lifecycle governance, ISO 13485 quality processes, IEC 62304 lifecycle practices, and engineering practices such as Specification-Driven Development (SDD) and Test-Driven Development (TDD) can work together in a regulated medical-device software program.

Evidence status: Anonymized portfolio reference. This is not a controlled QMS record or original project procedure.

1. Integrated Software Delivery Model

The software work was managed within a broader **hybrid Agile-Waterfall delivery framework**. Software teams worked with structured development and documentation practices while hardware development progressed through EVT, DVT, PVT, and mass-production readiness phases. The project-management role connected requirements, engineering execution, quality activities, traceability, testing, and delivery decisions.

Business / Clinical Need	Requirements & Scope	Software Planning
Specification / Design	Implementation	Unit & Integration Testing
Verification / Validation	Release Readiness	Deployment / Support

2. Quality & Lifecycle Governance

The project operated within a regulated medical-device environment. The documented project experience includes support for **ISO 13485 quality-management processes**, **IEC 62304 software lifecycle practices**, and **ISO 14971 risk-management activities**. QMS documentation and requirements traceability were managed through Confluence, with project-document review and approval performed in collaboration with the Compliance Officer.

The practical governance model linked:

Quality Management	ISO 13485 / QMS processes
Software Lifecycle	IEC 62304 lifecycle practices
Risk Management	ISO 14971 activities
Engineering Control	Specifications, development workflow, testing, version/document control
Project Governance	Scope, dependencies, reviews, risks, UAT, release and readiness

3. SDD & TDD as Engineering Practices

Specification-Driven Development (SDD) was used as an engineering practice to emphasize defining expected behavior and technical requirements before implementation. In a regulated environment, this supports clearer requirements, design decisions, documentation, and traceability.

Test-Driven Development (TDD) was used as an engineering practice to encourage tests to drive implementation and provide fast feedback during software development. TDD supports code-level verification; it does not replace system-level verification, validation, clinical validation, or the quality-system controls required for a medical-device program.

These practices should therefore be viewed as **engineering techniques inside the wider lifecycle**, rather than as substitutes for ISO 13485, IEC 62304, or project governance.

4. Traceability Across the Lifecycle

Requirement	Specification / Design	Implementation	Test Evidence	Release / Validation
What is needed	How it should behave	How it is built	Does it meet the defined behavior?	System ready for intended use?

5. Project Manager Contribution

The Senior Project Manager's role was not to replace the software engineers or compliance function. The contribution was to **establish and coordinate the delivery framework** in which engineering, quality, compliance, hardware, firmware, clinical, and operations work could progress together.

- Coordinate software, firmware, hardware, compliance, and operations dependencies.
- Maintain visibility of requirements, scope, documentation, traceability, risks, and delivery status.
- Facilitate Scrum practices for software teams within the hybrid delivery model.
- Coordinate reviews, demonstrations, testing/UAT, release readiness, and go-live activities.
- Support QMS documentation, audit readiness, and clinical-validation preparation.
- Promote structured development and documentation workflows using SDD/TDD where applicable to engineering work.

6. Evidence Boundary

This portfolio reference intentionally avoids proprietary source code, patient information, controlled QMS records, detailed clinical algorithms, confidential regulatory files, credentials, and restricted engineering records. It is designed to demonstrate the **delivery and engineering governance approach** without exposing confidential project material.

7. Key Takeaway

Business / Clinical Need → Requirements → Specification → Development → Testing → Verification & Validation → Release → Operations

The value of the model is the connection between **project management, regulated software lifecycle practices, engineering discipline, and evidence-based delivery**. SDD and TDD strengthen the engineering layer; ISO 13485, IEC 62304, ISO 14971, and project governance provide the wider control environment.

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